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Assistance Through Communication Technologies and Vehicles

The symbiosis between communication technologies and vehicles offers a priceless opportunity to improve assistance to people injured in traffic accidents, providing information about the incident to reduce the response time of emergency services. Determining the required human and material resources more accurately for each particular accident could significantly reduce the number of victims. This article presents a novel system prototype especially designed to detect and provide faster assistance for traffic accidents, thereby minimizing the consequences on the passengers' health. The proposed system requires each vehicle to be endowed with an onboard unit (OBU) responsible for detecting and reporting accident situations to an external control unit (CU) that estimates its severity, allocating the necessary resources for the rescue operation. The development of a prototype based on off-the-shelf devices and its validation at the Applus+ IDIADA Automotive Research Corporation facilities show that our system could notably reduce the time needed to alert and deploy the emergency services after an accident takes place.



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AUTOMATIC ACCIDENT DETECTION

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During the last few decades, the total number of vehicles around the world has experienced a remarkable growth, making the traffic density higher and increasing the attention of drivers. The immediate effect of this situation is the dramatic increase of traffic accidents on the road, representing a serious problem in most countries. For example, 2,478 people died on Spanish roads in 2010, which means one death for every 18,551 inhabitants [1].

Numerous efforts have been undertaken by automobile manufacturers to reduce road casualties, mainly focused on both active and passive safety systems. These initiatives have managed to increase traffic safety, achieving a reasonable reduction of road deaths. However, accidents can still occur, and a quicker response from emergency services could significantly decrease both the number of injured and dead passengers as well as the impact and severity of such accidents.

The European Commission is currently funding several projects under the i2010 Intelligent Car Initiative, which promotes several efforts toward new safety systems. Cooperative systems using vehicle-to-vehicle (V2V) communications are now considered necessary to accomplish these objectives and will play an increasing role in the intelligent transportation systems (ITS) area. Most ITS applications, such as road safety, fleet management, and navigation, will rely on information and communication technologies between the vehicle and the roadside infrastructure (V2I) or between vehicles (V2V).

In this article, we present a prototype architecture called *e-NOTIFY*, a novel proposal designed to improve the chances of survival for passengers involved in car accidents. The proposed system offers automated detection, reports, and assistance to passengers involved in road accidents by exploiting the capabilities offered by vehicular communication technologies. Our proposal does not focus on reducing the number of accidents but on improving postcollision assistance with fast and efficient management of the available emergency resources, thereby increasing the chances of recovery and survival for people injured in traffic accidents.

Motivation

When a traffic accident takes place, assisting injured passengers as soon as possible is crucial to minimize the negative effects on their health. Mortality from traffic accidents can be classified into three phases [2]:

- Phase 1 involves casualties in the first few minutes or seconds after an accident (about 10% of all deaths).
- Phase 2 is the first hour after the accident, the so-called golden hour, has the highest mortality (75% of all deaths) and is the phase during which the highest death rate can be avoided by proper initial health care.
- Phase 3 happens days or weeks after the traumatic incident, has 15% of mortality, and takes hard work and a high amount of resources to reduce mortality.

As can be observed, the phase where more benefits can be achieved by reducing rescue response time is the second one. A fast and efficient rescue operation during the hour after a traffic accident significantly increases the probability of survival of the injured and reduces the injury severity.

For a noticeable reduction in rescue time, two major steps must be taken: 1) fast and accurate accident detection and reporting to an appropriate public safety answering point (PSAP) and 2) fast and efficient evacuation of occupants trapped inside a vehicle. The first of these objectives can be accomplished by using telecommunication technologies incorporated into the automotive world. There have been many advances in the development of communication between vehicles (V2V technologies), also known as *vehicular ad hoc networks* (VANETs) [3], offering support for cooperative security applications between vehicles. In fact, the 802.11p working group recently approved the IEEE 802.11p standard [4], providing a viable solution for intervehicular security applications. This technology already has been studied to increase traffic safety in dangerous areas such as intersections [5].

However, accomplishing the second goal is becoming harder every year. Studies conducted by a German automobile club, Allgemeiner Deutscher Automobil Club (ADAC) [6], have proved that the rescue operation of injured people from a vehicle is long when the vehicle is a newer model. This effect is clearly visible in Figure 1, where the impact of the year of manufacture of the vehicle in relation to rescue speed is shown. The increase of security equipment that makes vehicles safer also implies more complexity for the emergency teams. From a golden-hour perspective, this is a serious threat to the successful rescue of injured persons.

Increasing the amount of information available about the accident and the vehicle involved could definitely contribute to the second goal [7]. In fact, the effectiveness of

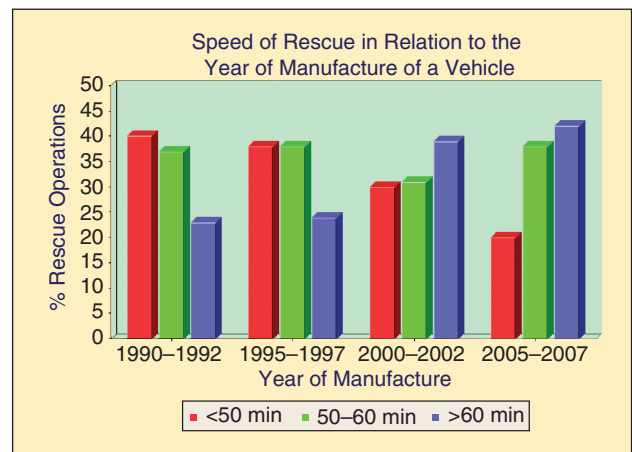


FIGURE 1 Impact of the year of manufacture of the vehicle in relation to rescue speed. Data provided by the ADAC [6].

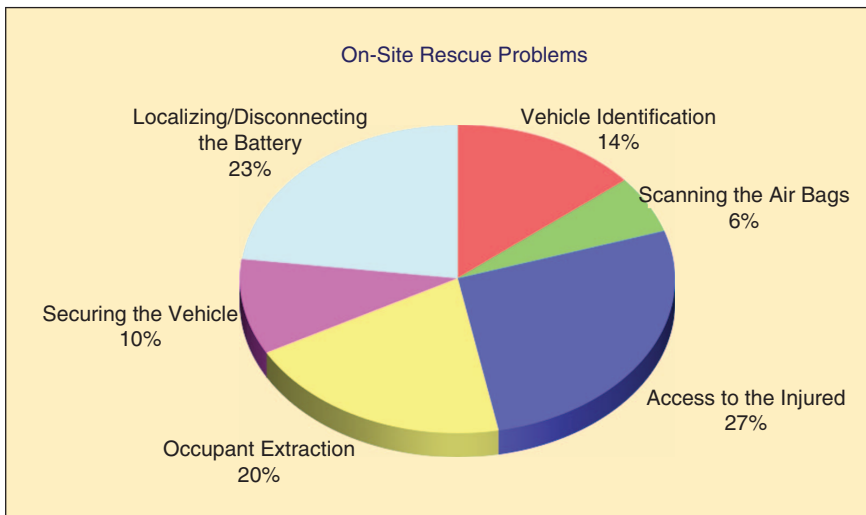


FIGURE 2 Main rescue problems at an accident site. Data provided by the ADAC [6].

the assistance to passengers involved in a traffic accident could be significantly improved if emergency services had available relevant information on the conditions under which the accident happened before moving to the area of the accident. As shown in Figure 2, up to 63% of all problems that rescue teams can find when facing an accident could be reduced by providing emergency services with additional information. This extra information, obtained from sensors inside the vehicle, could be used to estimate the severity of the injuries of the occupants. Also, having more information would allow emergency providers to determine the optimal set of human and material resources to be sent to the accident location, with a consequent assistance quality improvement.

Related Projects

A number of research projects headed by different research institutes and car manufacturers around the world have been focusing on intervehicle communication systems. Some of the larger projects related to the e-NOTIFY system are listed below:

- **COMeSafety2** [8]: The COMeSafety2 project proposal aims at coordinating the activities toward the realization of cooperative systems on European roads, focusing on all issues related to V2V and V2I communications. Its main goal consists of developing a European set of standards to support wide implementation and deployment of cooperative ITS.
- **eCall** [9]: The eCall system has been designed to improve transportation safety, providing rapid assistance to people involved in a collision anywhere in the European Union. A collision activates an emergency voice call to be established via the cellular network to local emergency agencies. In addition, eCall transmits a minimum set of data (MSD), including key information about the accident, such as time,

location, and vehicle description. It is supposed that the eCall system will become operative by 2015.

- **OnStar** [10]: The OnStar is an in-vehicle safety and security system created by General Motors (GM) for on-road assistance, which resembles the European eCall project. A collision activates an emergency voice call to report key information about the accident.
- **Campus Vehicular Test Bed** [11]: The Campus Vehicular Test Bed is an effort by the University of California, Los Angeles to deploy an open test bed that integrates ad hoc V2V commu-

nications and a wireless mesh to offer support for communications.

The most similar projects to e-NOTIFY are the eCall and OnStar projects, which are expected to be manually activated or using the in-vehicle sensors for air-bag deployment. However, our proposal goes one step beyond their aims. We are developing an autonomous intelligent system that allows automatically adapting the required rescue resources to each particular accident, allowing the rescue staff to work far more efficiently and reducing the time associated to their tasks.

e-NOTIFY System: Architecture Overview

Figure 3 presents the basic structure of an e-NOTIFY system. The proposed system consists of several components with different functions. First, the vehicles should incorporate an OBU responsible for detecting accidents and communicating information about dangerous situations. Next, the notification of the detected accidents is made through a combination of both V2V and V2I communications. Finally, the destination of the information is the CU that will handle the warning notification, estimating the severity of the accident and communicating the incident to the appropriate emergency services.

The OBU definition is of utmost importance for the proposed system. This device must be technically and economically feasible, as its adoption in a wide range of vehicles could increase in the near future. In addition, this system should be open to future software updates. Although the design of the hardware to be included in vehicles initially consisted of special-purpose systems, this trend is heading toward general-purpose systems because of the constant inclusion of new services.

The information exchange between the OBUs and the CU is made through the Internet, either through vehicles providing Internet access [via Universal Mobile

Telecommunications System (UMTS), for example] or by reaching infrastructure units [roadside units (RSU)] that provide this service. If the vehicle does not get direct access to the CU on its own, it can generate messages to be broadcast by nearby vehicles until they reach one of the aforementioned communication paths. These messages, when disseminated among the vehicles in the area where the accident took place, also serve the purpose of alerting drivers traveling to the accident area about the state of the affected vehicle and its possible interference on the normal traffic flow. An efficient warning message dissemination protocol should exploit the street and buildings layout of the surrounding area to carefully select the most appropriate forwarding node for the message [12]. This allows maximizing the percentage of informed vehicles and reducing the time elapsed between the accident occurrence and its actual notification to each specific vehicle (i.e., the warning notification time) while simultaneously reducing the amount of traffic generated in the wireless channel.

The goal of our proposal is to provide an architecture that allows 1) direct V2V involved in the accident, 2) automatic sending of a data file containing important information about the incident to the CU, and 3) a preliminary and automatic assessment of the damage to the vehicle and its occupants, based on the information received from the involved vehicles, and a database of accident reports. According to the reported information and the preliminary accident estimation, the system will alert the required rescue resources to optimize accident assistance.

OBU Design

The main objective of the e-NOTIFY OBU lies in obtaining the available information from sensors inside the vehicle to determine when a dangerous situation occurs, and reporting that situation to the nearest CU, as well as to other nearby vehicles that may be affected.

OBU Internal Structure

Figure 4 shows the e-NOTIFY OBU system, which relies on the interaction between sensors, the data acquisition unit (DAU), the processing unit, and wireless interfaces:

- *In-vehicle sensors:* In-vehicle sensors are required to detect accidents and provide information about its causes. Accessing the data from in-vehicle sensors is

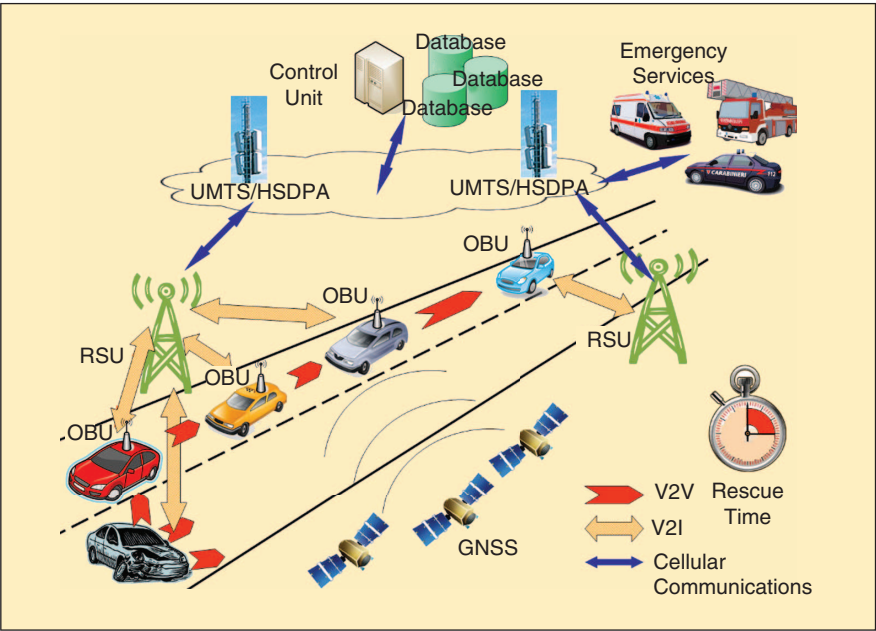


FIGURE 3 e-NOTIFY architecture based on a combination of V2V and V2I communications.

possible nowadays using the onboard diagnostics standard interface [13], which serves as an entry point to the vehicle's internal bus. This standard is mandatory in Europe and the United States since 2001. This encompasses the majority of the vehicles of the current automotive park, and the percentage of compatible vehicles will keep growing as very old vehicles are replaced by new ones.

- *DAU:* DAU is responsible for periodically collecting data from the different sensors available in the vehicle (air-bag triggers, speed, and fuel levels), converting them to a common format, and providing the collected data set to the OBU processing unit.
- *OBU processing unit:* OBU processing unit is in charge of processing the data from sensors, determining whether an accident occurred, and notifying dangerous situations to nearby vehicles and to the CU. The information from the DAU is gathered, interpreted,

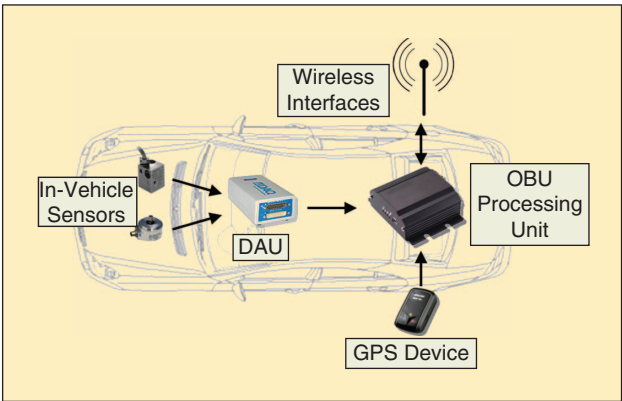


FIGURE 4 OBU structure diagram.

NUMEROUS EFFORTS HAVE BEEN UNDERTAKEN BY AUTOMOBILE MANUFACTURERS TO REDUCE ROAD CASUALTIES, MAINLY FOCUSED ON BOTH ACTIVE AND PASSIVE SAFETY SYSTEMS.

and used to determine the vehicle's current status. This unit must have access to a positioning device [such as a global positioning system (GPS) receiver] and to different wireless interfaces, thereby enabling V2V and the remote control center.

Accident Detection Algorithm

The first goal of the OBU consists of determining when a dangerous accident occurs. In the traffic accidents domain, there are two main events that could cause severe damage to the passengers in a vehicle: rollovers (overturns) and strong impacts. We are currently working with the Applus+ IDIADA Automobile Research Corporation [14] to develop a realistic accident detection algorithm based on information that characterizes different types of accidents.

Crash tests held by IDIADA collect a huge amount of information about the collision (10,000 samples/s), which is unfeasible to be handled in real time, thus it must be processed offline after an accident. Nevertheless, for a really useful system, data must be processed at the moment of the accident to reduce the assistance time and the effects of collision on the passengers. Moreover, the equipment used by IDIADA to record all this information is not affordable in a standard vehicle. Therefore, our detection system should be based on an affordable online

system but still accurate to detect when an accident occurs. So, e-NOTIFY OBUs use a reduced sampling frequency compared with the configuration under IDIADA tests. The new sampling frequency is selected so that it is possible to handle it in real time, while being precise enough to classify the different types of accident pulses. Experiments showed that about 100 measurements per second are adequate to achieve a tradeoff between accuracy and real-time processing.

When trying to detect an accident, a rollover in a vehicle is quite simple to recognize using a horizontal tilt sensor, since measurements deviating more than 90° from the horizontal or a constant value over 45° (partial rollover) indicate that the vehicle overturned and needs to be rescued.

The interpretation of acceleration values is more complicated. The straightforward approach to classify collisions would consist of defining a series of acceleration thresholds. Nevertheless, this simple method is not valid for all tested situations, as shown in Figure 5. The graph contains different pulses corresponding to front crashes with different severities. As shown, the peak acceleration recorded in the minor accident exceeds the maximum value registered in the severe collision, although the duration of the pulse is much smaller. Therefore, it is clear that using simple acceleration thresholds to distinguish the acceleration pulses is not enough, and both their amplitude and duration should be considered to better estimate the severity of accidents.

To take into account both the amplitude and duration of the pulse, the e-NOTIFY system uses the area that the pulse forms with the time axis, which can be obtained applying the integral of the function. Therefore, the integral value of the function is used to classify acceleration pulses and determine the preliminary severity of an impact.

The DAU is configured to collect as much data as possible from the sensors during the intervals between sampling, calculating the average value measured during the period before sending it to the process unit. Hence, the most appropriate method for approximating the integral value during an interval will be the rectangle rule, calculated using

$$\int_{x_0}^{x_n} f(x)dx \approx \sum_{i=1}^n (x_i - x_{i-1}) \cdot f_{\text{avg}}[x_{i-1}, x_i],$$

where n is the number of intervals recorded and f_{avg} is the average value of the function $f(x)$ in the interval $[x_{i-1}, x_i]$. The integration of the acceleration function starts when a value over 6G (for front collisions) or 3G (for side and rear-end collisions) is detected. These values were found and adjusted by using the historical information about crash tests provided by Applus+ IDIADA.

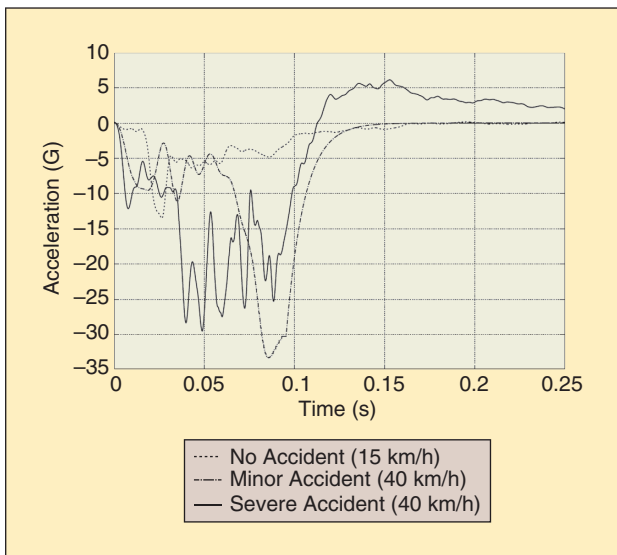


FIGURE 5 Acceleration pulses for different front-crash ratings. Data provided by Applus+ IDIADA Corporation [14].

**WHEN A TRAFFIC ACCIDENT TAKES PLACE,
ASSISTING INJURED PASSENGERS AS SOON
AS POSSIBLE IS CRUCIAL TO MINIMIZE THE
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OBU Design Under the OSGi Environment

The e-NOTIFY OBUs make use of the open services gateway initiative (OSGi) standard [15], which enables the development of applications (in the form of bundles or modules for deployment) that can be installed, started, stopped, updated, and uninstalled remotely without rebooting the system. Many car manufacturers have included the OSGi specification in their global system for telematics (GST) specification. Since OSGi is becoming the de facto standard for vehicular network systems, it has been chosen to be implemented in the e-NOTIFY system.

In the OSGi environment, a bundle is an application packaged in a Java archive (JAR), which is deployed in an OSGi platform. Therefore, the applications to be executed by the OBUs must be programmed in the Java language and packaged in a JAR file that allows deployment as a module in the system. The inclusion of new services may

be carried out using a similar procedure, obtaining a highly scalable and updateable architecture.

CU Design

The CU is associated to the response center in charge of receiving notifications of accidents from the OBUs installed in vehicles. In particular, the CU is responsible for dealing with warning messages, retrieving information from them, and notifying the emergency services about the conditions under which the accident occurred.

CU Internal Structure

Figure 6 shows the modules included in the CU to achieve all of its objectives within the e-NOTIFY system.

- **Reception/interpretation module:** The first step for the CU is to receive a warning message from a collided vehicle, and so there must be a module waiting for the arrival of messages and obtaining their different fields.
- **Accident severity estimation module:** When a new accident notification is received, this module will determine how serious the collision was and the severity of the injuries of passengers.
- **Resource assignment module:** After deciding the severity of the accident, an additional module is used to define resource sets adapted to the specific situation.

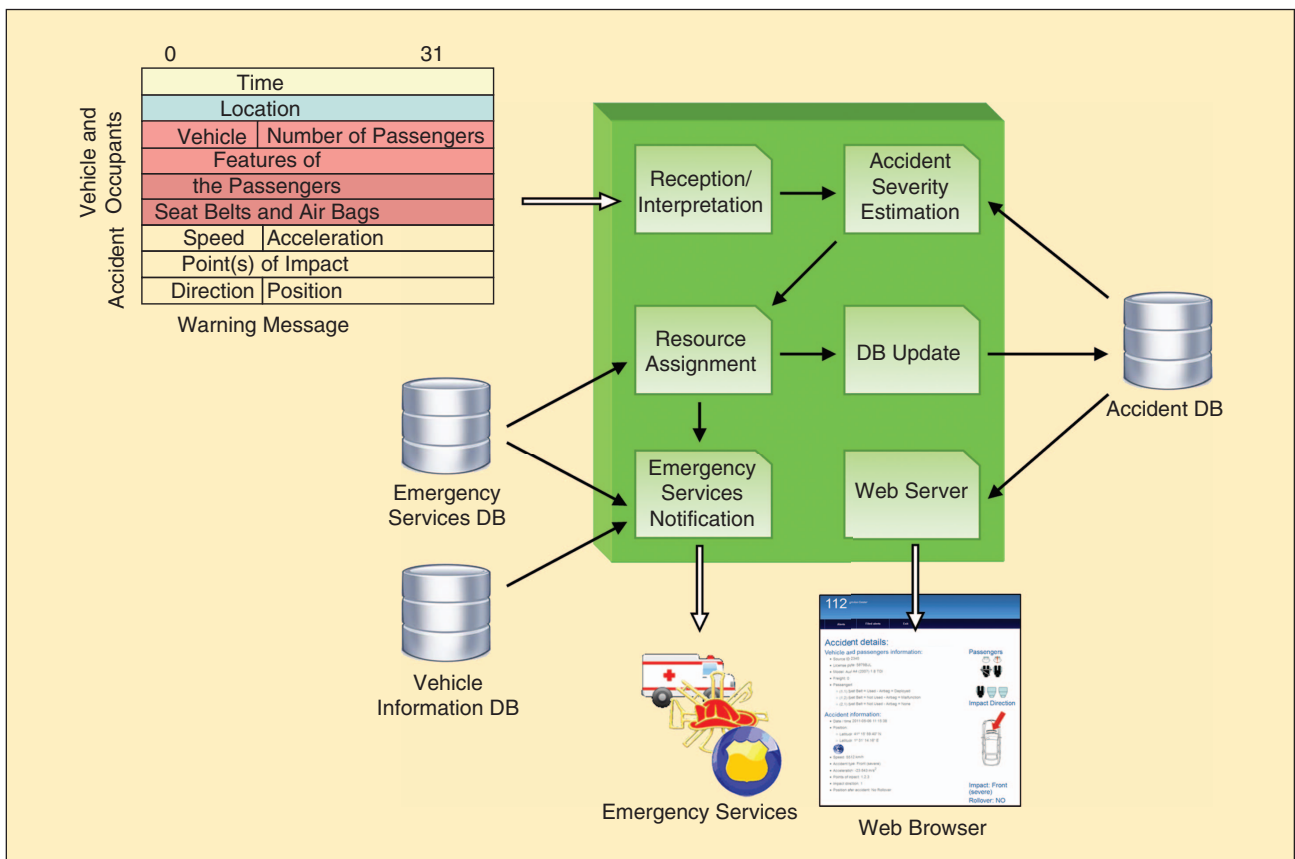


FIGURE 6 ACU modular structure.

THE FIRST GOAL OF THE OBU CONSISTS OF DETERMINING WHEN A DANGEROUS ACCIDENT OCCURS.

- **Database update module:** The data collected from the notified accident are stored into an existing database of previous accidents, increasing the knowledge about the accident domain.
- **Web server module:** The CU incorporates a Web server to allow easy visualization of the historical information recorded and the current accident situations requiring assistance. A Web interface was chosen to increase user friendliness and interoperability.

- **Emergency services notification module:** When the information has been correctly managed, the notification module sends messages to the emergency services, including all the information collected, the estimated severity, the recommended set of resources, as well as additional information about the vehicles involved in the collision (for preliminary planning of the rescue operation). The information about vehicles consists of standard rescue sheets, which highlight the important or dangerous parts of a specific vehicle that should be taken into account during a rescue operation, such as batteries and fuel tanks, as shown in Figure 7.

The CU makes use of three different databases to handle accident notifications. The accident database contains historical data of past accidents detected by the system, and it is used to build the estimation models that predict the severity of new accidents. The vehicle information database provides, for each vehicle, available information about rescue procedures and dangerous components inside each car. Finally, the emergency services database includes information about the rescue services available in the area of influence of the CU to determine the emergency services needed for each specific accident.

Accident Severity Estimation

After receiving an accident notification, the CU must determine the severity of the traffic accident to adjust the available resources to each situation. In particular, the questions that must be answered to obtain useful information are 1) How damaged are the vehicles involved in the accident? and 2) How severe are the injuries suffered by the passengers? The first question will determine the need of machinery, such as cranes to restore normal traffic flow, or the likability for the vehicle to catch fire and cause additional dangerous situations. The second one is related to the health equipment and the vehicles necessary to increase the probability of survival of the vehicle occupants.

Developing a useful algorithm to estimate accident severity needs

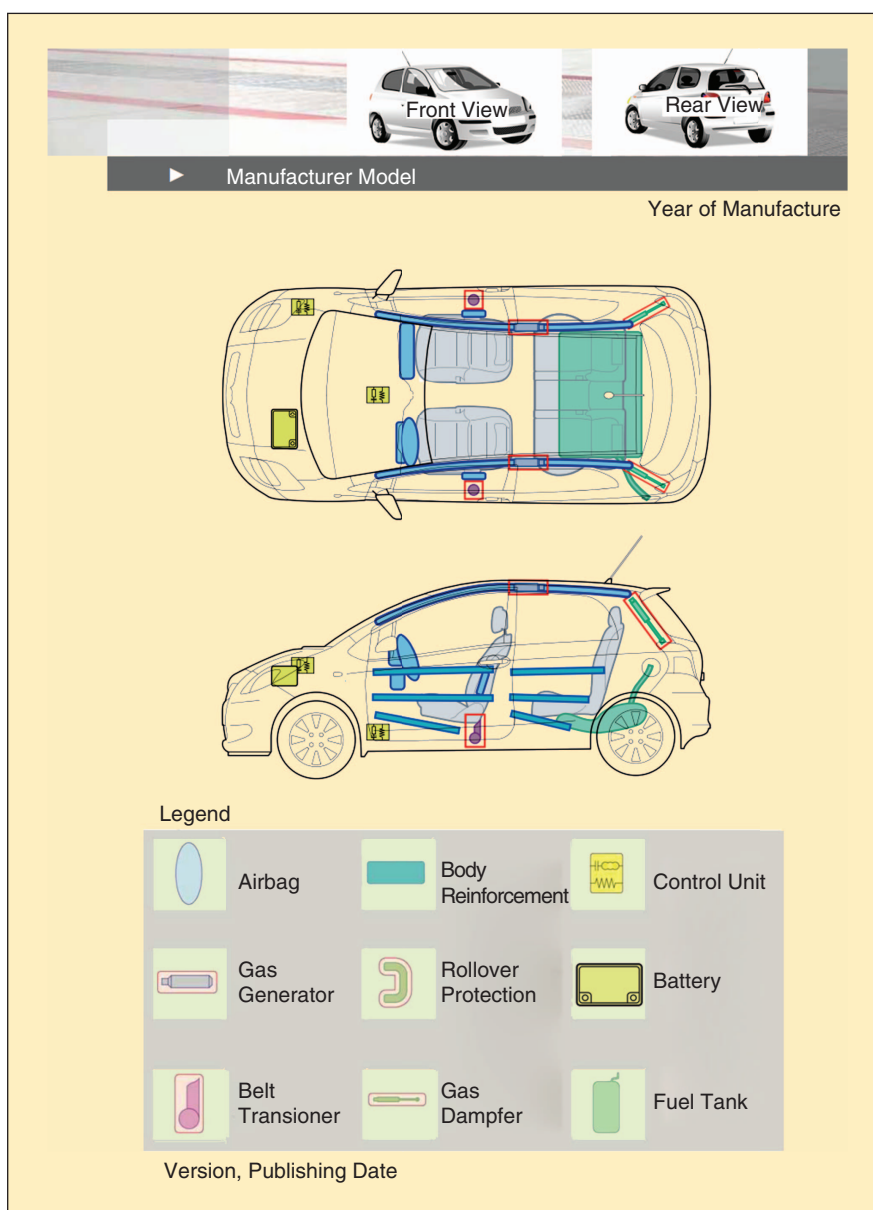


FIGURE 7 An example of standard rescue sheet.

historical data to ensure that the criteria used are suitable and realistic. The National Highway Traffic Safety Administration maintains the general estimates system (GES) [16], a database with information about traffic accidents that began its operation in 1988. The data for this database is obtained from a sample of police accident reports (PARs) collected all over the U.S. roads. This database includes information about the crash characteristics and environmental conditions at the time of the accident, vehicles and drivers involved in the crash, and the people involved in the collision.

Using the data contained in the GES database, the damage in vehicles is classified into three categories, minor, moderate, and severe damage, depending on whether the vehicle can be driven safely or not. Focusing on passenger injuries, three different classes are used to determine their severity: no injury, nonincapacitating injury, and incapacitating or fatal injury.

The data-mining classification algorithms used to estimate the accident severity were built using the Weka open-source data mining package [17]. By using data mining classification algorithms based on Bayesian networks and the K2 search algorithm [18], trained with the values from previous accidents in the GES database, we are able to generate classification models that correctly estimate the accident severity in more than 75% of the cases [19]. Bayesian models have an advantage of introducing a degree of complexity that does not depend on the number of instances used during the training phase of the algorithm, presenting a reduced number of parameters for the model definition. Therefore, these estimations may be used to adapt the resources to the conditions of the accident.

Prototype Implementation and Validation

A prototype for the e-NOTIFY system was built using off-the-shelf devices, allowing fast development and reduced cost.

OBU Prototype

The DAU in the initial prototype is built using an ARM microcontroller programmed to periodically collect data from in-vehicle sensors. Basically, these sensors are accelerometers and gyroscopes that indicate the severity of the impacts received by the automobile or the occurrence of a rollover that might endanger the integrity of the occupants. Communication between the microcontroller and the processing unit is done by sending user datagram protocol packets through an Ethernet interface.

The OBU processing unit in our prototype is a general-purpose Asus Eee PC netbook, equipped with a solid-state disk (SSD) to minimize the possibility of damage due to impact in crash tests. The vehicle position and speed are obtained using a GPS device accessible using Bluetooth.

THE DAU IS CONFIGURED TO COLLECT AS MUCH DATA AS POSSIBLE FROM THE SENSORS DURING THE INTERVALS BETWEEN SAMPLING, CALCULATING THE AVERAGE VALUE MEASURED DURING THE PERIOD BEFORE SENDING IT TO THE PROCESS UNIT.

CU Prototype

The CU prototype for the initial tests was built using common software components, allowing fast prototyping with less cost. The reception/interpretation module was implemented using the Java programming language. This module acts as a concurrent server, creating different execution threads to handle each message received, which allows exploiting multiprocessor or multicomputer architectures.

Databases are managed using the MySQL relational database management system. MySQL was selected because of its scalability and easy integration with the rest of components of the CU.

The Web server for the visualization module is Apache. To support dynamic content, we use the hypertext preprocessor technology, which is easily integrated into Apache. By combining these technologies and MySQL, users can visually check the system status, as shown in Figure 8.

Prototype Validation

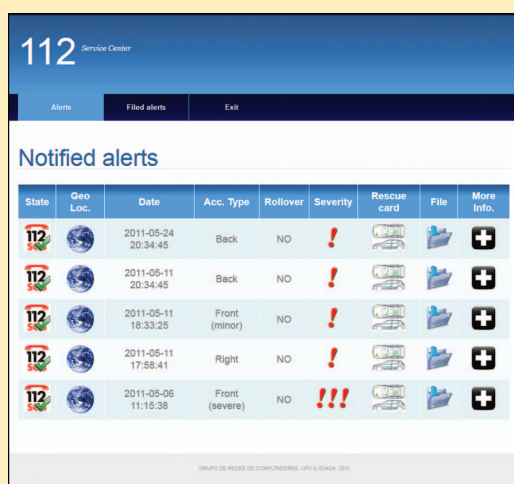
The prototype was validated at the Applus+ IDIADA [14] Passive Security Department facilities in Santa Oliva (Tarragona, Spain). These facilities house one of the most sophisticated crash test laboratories in the world and constitute an official center for approval under the European New Car Assessment Programme program.

Due to the cost of using real vehicles in the collision experiments, the e-NOTIFY prototype tests were carried out using a platform (known as *sled*) that moves on rails to collide against a series of metal bars that simulate the deformation suffered by a vehicle body to absorb the impact. The speed of the stroke and the configuration of bars used in the test determine, respectively, the kind of accident detected and the segment the simulated vehicle belongs to (family car and off road). Tested speeds are determined by European standards and vary from 10 to 64 km/h to represent different accident severities.

Figure 9 shows the sled used in the tests. Validation experiments consisted of front, side, and rear-end crash tests, accounting for both accident and no-accident situations. The classification of the severity of the collision is dictated by the parameters used in Applus+ IDIADA in automotive standard tests. The specific tests carried out during the validation phase appear in Table 1.

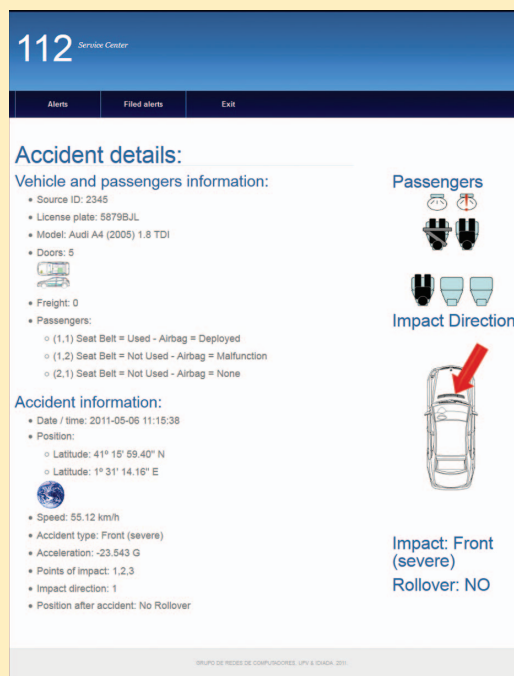
THE CU IS ASSOCIATED TO THE RESPONSE CENTER IN CHARGE OF RECEIVING NOTIFICATIONS OF ACCIDENTS FROM THE OBUS INSTALLED IN VEHICLES.

The test system included an external computer receiving regular information from the sled (via a wireless network) of the measurements recorded by the OBU to ensure the proper behavior of the sensor reading module,



State	Geo Loc.	Date	Acc. Type	Rollover	Severity	Rescue card	File	More Info.
112		2011-05-24 20:34:45	Back	NO	!			
112		2011-05-11 20:34:45	Back	NO	!			
112		2011-05-11 18:33:25	Front (minor)	NO	!			
112		2011-05-11 17:58:41	Right	NO	!			
112		2011-05-06 11:15:38	Front (severe)	NO	!!!			

(a)



Accident details:	
Vehicle and passengers information: - Source ID: 2345 - License plate: 5879BJL - Model: Audi A4 (2005) 1.8 TDI - Doors: 5 - Freight: 0 - Passengers: (1,1) Seat Belt = Used - Airbag = Deployed (1,2) Seat Belt = Not Used - Airbag = Malfunction (2,1) Seat Belt = Not Used - Airbag = None	Passengers Impact Direction
Accident information: - Date / time: 2011-05-06 11:15:38 - Position: - Latitude: 41° 15' 59.40" N - Longitude: 1° 31' 14.16" E - Speed: 55.12 km/h - Accident type: Front (severe) - Acceleration: -23.543 G - Points of impact: 1,2,3 - Impact direction: 1 - Position after accident: No Rollover	Impact: Front (severe) Rollover: NO

(b)

FIGURE 8 Web interface screenshots with information about notified accidents.

along with a CU in charge of receiving alert messages and applying the corresponding algorithms. The real trials were carried out with two different objectives, proving that the OBU prototype was solid enough to resist a dangerous impact; thus, it could continue to work after the accident and ensures proper function of the system under a realistic crash situation.

All the tests produced positive results since the OBU did not suffer from noticeable damage even during the strongest impacts. The experiment helped to show that the OBU was able to correctly detect both the magnitude and direction of the impact. Figure 10 summarizes how acceleration pulses were handled by the e-NOTIFY system in two of the experiments. As shown, using a reduced sampling frequency, a similar pulse shape is obtained with less than 10% variation in the integral value (approximated by the sum of smaller rectangle areas) compared to the area obtained using the highest sampling frequency.

In addition, the OBU generated an appropriate warning message from the sensor data and sent it using the UMTS technology to the CU in all accident configurations, as shown in Figure 11. The latter properly processed the accident details, generating a correct estimation of the severity of the accident.

Conclusions

In this article, we presented an e-NOTIFY system, which allows fast detection of traffic accidents, improving the assistance to injured passengers by reducing the response time of emergency services through the efficient communication of relevant information about the accident using a combination of V2V and V2I communications. The proposed system requires installing OBUs in the vehicles, in charge of detecting accidents and notifying them to an external CU, which will estimate the severity of the accident and inform the appropriate emergency services about the incident. This architecture replaces the current mechanisms for notification of accidents based on

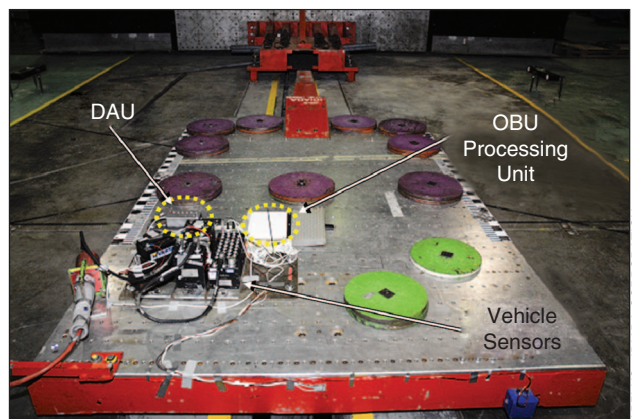


FIGURE 9 Sled with the e-NOTIFY prototype installed before a crash detection test.

TABLE 1 Validation tests carried out on the e-NOTIFY system.

Accident Type	Vehicle Segment	Accident Severity	Pulse Duration	Acceleration
Front accident	Large family car	Severe accident	110 ms	23–28 G
	Large MPV	Minor accident	100 ms	15–21 G
	Small family car	No accident	110 ms	4–9 G
Side accident	Small offroad 4×4	Accident	90 ms	14–21 G
	Supermini	No accident	90 ms	3–6 G
Rear-end accident	Small MPV	Accident	110 ms	5–7 G
	Supermini	No accident	70 ms	2–6 G

witnesses, who may provide incomplete or incorrect information after a long time. The development of a low-cost prototype shows that it is feasible to massively incorporate this system in existing vehicles. We validated our prototype at the Passive Security Department of Applus+ IDIADA Corporation, and showed how it can successfully detect traffic accidents, reporting all the detailed information to a control alert system on time. Future work in this

area includes deploying the system in a real environment with the OBUs installed in real vehicles to check the system behavior when moving at high speeds.

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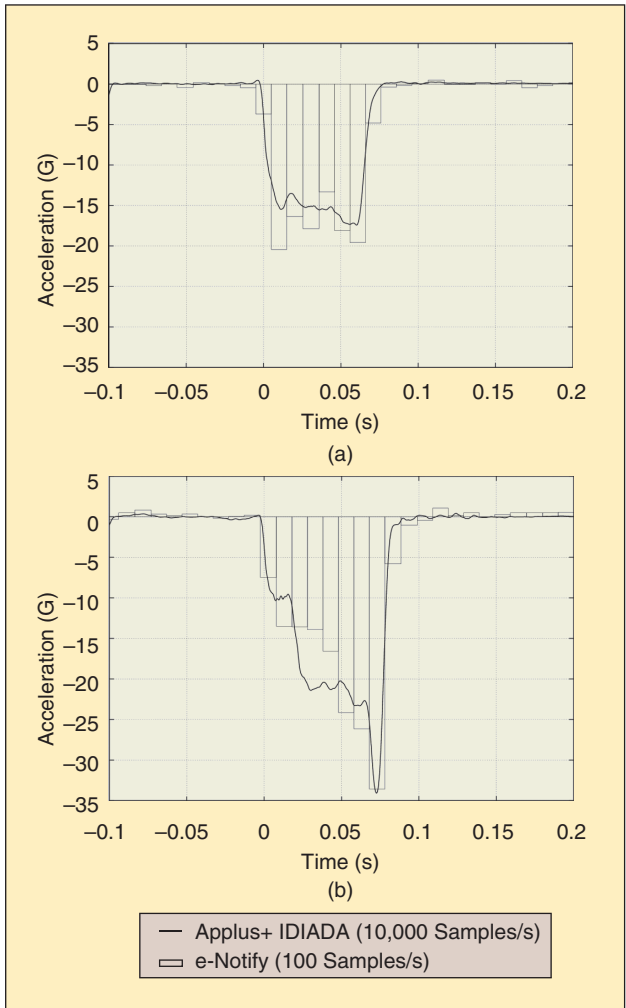


FIGURE 10 Acceleration pulses collected by Applus+ IDIADA compared to the samples obtained by the e-NOTIFY system in the same experiments: (a) front minor accident and (b) front severe accident.

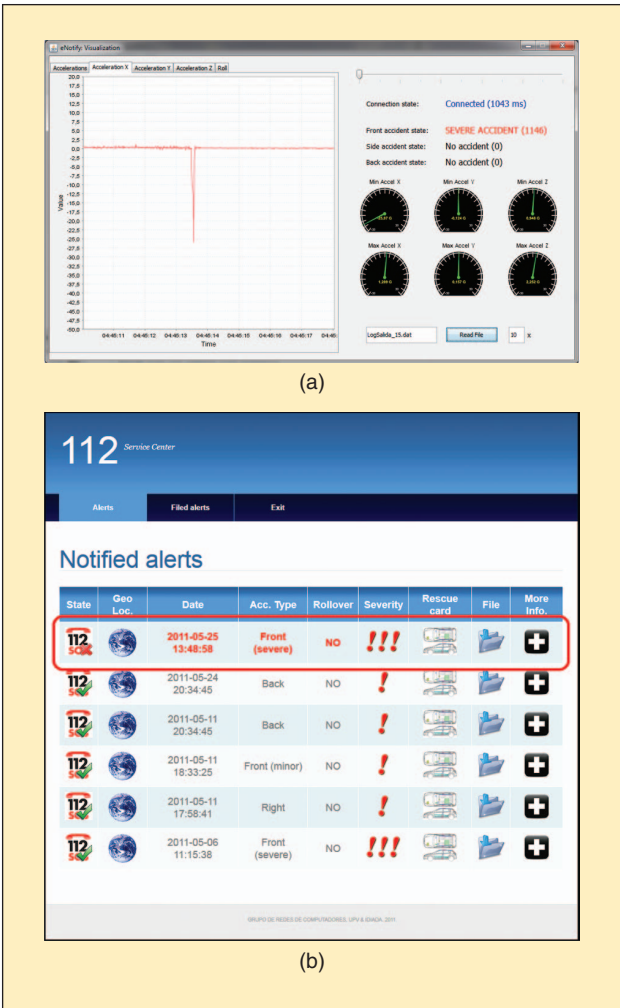


FIGURE 11 Images of the crash test results. (a) Accident pulse recorded by the OBU and (b) the same accident notified and received by the CU.

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